

CONTEXT-CONDITIONED SEQUENTIAL KEYPOINT DENSITY ESTIMATION FOR TRAFFIC HAZARD DETECTION

PROGRESS REPORT

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ABSTRACT

The Sequential Keypoint Density Estimator (SeeKer / SKDE) is a recent ICCV 2025 baseline for skeleton-based video anomaly detection that models the joint density of a pedestrian’s 2D keypoint trajectory through an autoregressive factorization. Despite its simplicity, SeeKer matches or surpasses much heavier graph-convolutional and diffusion-based competitors. However, it is *environmentally blind*: each skeleton is treated in complete isolation from the surrounding scene. In real-world surveillance — and especially in traffic-hazard scenarios — a pedestrian’s motion is rarely abnormal in absolute terms; it is the *interaction* with vehicles, bicycles, or restricted zones that defines the hazard. We propose **CCSKDE** (Context-Conditioned SKDE), an extension that injects a lightweight environmental context vector C_t — derived from off-the-shelf YOLOv11 detections of non-pedestrian scene elements — directly into the autoregressive factorization. We evaluate on the ShanghaiTech Campus dataset, which natively contains pedestrian–vehicle interactions. This document reports our problem framing, related work, the proposed method, the experimental protocol, and preliminary results from reproducing the SeeKer baseline.

1 PROBLEM DEFINITION

Video anomaly detection (VAD) is the task of identifying frames or events in a video stream whose appearance, motion, or interaction patterns deviate significantly from those observed in a training set of *normal* videos. Modern VAD pipelines increasingly operate not on raw RGB pixels but on intermediate, privacy-preserving representations such as 2D human skeletons extracted by an off-the-shelf pose estimator. Skeleton-based VAD has three practical advantages: (i) the input is much lower-dimensional than RGB, (ii) models become invariant to appearance and lighting biases that often plague RGB-based detectors, and (iii) no facial or identifying pixels are retained, which is desirable for surveillance applications.

Within this family, the *Sequential Keypoint Density Estimator* (SKDE / SeeKer) of Delić et al. (2025) is a particularly elegant baseline. It treats a pedestrian’s pose trajectory as a high-dimensional time series and estimates its joint density via an autoregressive factorization at the keypoint level: each individual 2D keypoint is predicted as a Gaussian conditioned on all previously generated keypoints in the same and earlier frames. Anomalies are detected by ranking observed test sequences by their log-likelihood under the model. Despite using only a small masked fully-connected (MADE) network (Germain et al., 2015), SeeKer matches or exceeds heavier flow-based and graph-based competitors on standard benchmarks (Hirschorn & Avidan, 2023; Liu et al., 2018).

Limitation we address. SeeKer’s autoregressive distribution is conditioned only on the pedestrian’s own past keypoints; it never observes *what else is in the scene*. This is a fundamental in-

formation bottleneck: a pedestrian walking at a constant velocity in a calm campus square and the same pedestrian walking at the same velocity directly into the path of a moving vehicle yield identical SeeKer likelihoods. In surveillance, however, the latter is precisely the event we need to flag. The aim of this project is to extend SeeKer with a minimal, mathematically faithful conditioning mechanism on a per-frame *environmental context vector* C_t , so that the predicted keypoint distribution can sharpen (smaller covariance) when the pedestrian is in a high-risk spatial configuration.

Challenges. The problem is non-trivial for four reasons. (i) **Unsupervised conditioning:** VAD is one-class, so we must learn what contextual configurations are normal without any hazard labels. (ii) **Preserving the autoregressive factorization:** naively concatenating C_t to MADE’s input breaks the carefully designed keypoint mask, so the conditioning must be injected as an unconditional shortcut while the keypoint-to-keypoint causality is left intact. (iii) **Generalization across scenes:** ShanghaiTech contains 13 different camera viewpoints, so C_t must be a scene-scale-invariant function of pedestrian–object geometry rather than absolute image coordinates. (iv) **Compute envelope:** the project must run in a single-semester timeline on commodity hardware (Apple Silicon MPS in our case), ruling out heavy dynamic-graph or transformer backbones. We note that ShanghaiTech is the only widely-used VAD benchmark with mixed pedestrian/vehicle interactions; UBnormal (Ac-sintoae et al., 2022) is fully synthetic, so our context contribution is evaluated primarily on ShanghaiTech.

2 RELATED WORK

Skeleton-based video anomaly detection. Pose-based approaches have become the dominant paradigm for VAD on person-centric datasets, since they sidestep RGB-level confounders. Morais et al. (2019) introduced one of the first skeleton-only VAD pipelines using a recurrent message-passing network. Markovitz et al. (2020) cast normal motion modeling as soft-clustering of spatio-temporal graph embeddings. STG-NF (Hirschorn & Avidan, 2023) fits a normalizing flow over spatio-temporal graphs of joints and was the state of the art on ShanghaiTech-HR until SeeKer. SeeKer (Delić et al., 2025) departs from these heavier architectures and demonstrates that an autoregressive masked fully-connected network is in fact a strong — and previously overlooked — baseline. Our work builds directly on SeeKer.

Density estimation backbones. SeeKer’s core is MADE (Germain et al., 2015), a masked fully-connected autoencoder that performs autoregressive density estimation in a single forward pass. Related neural density estimators (RealNVP (Dinh et al., 2017), Glow (Kingma & Dhariwal, 2018)) require multiple coupling layers but produce exact log-likelihoods; STG-NF exploits this for VAD scoring. SeeKer shows the simpler masked-FC formulation is sufficient and substantially cheaper.

Context-aware anomaly detection. A separate strand of VAD literature explicitly injects scene/object information into the anomaly scorer. Ionescu et al. (2019) score per-object appearance and motion features using one-class SVMs, and Georgescu et al. (2021) use background-aware self-supervised pretexts. The HSC (Sun & Gong, 2023) and SSMTL++ (Georgescu et al., 2022) frameworks combine multiple self-supervised signals including object-level cues. To our knowledge, no prior work has injected an explicit, per-frame proximity-to-non-pedestrian context vector into an autoregressive keypoint density estimator; this is the gap our work addresses.

Object detection, datasets and pose extraction. We use YOLOv11 (Ultralytics, 2024) for one-time offline detection of non-pedestrian scene elements; its bounding boxes are aggregated into C_t as described in Section 3.2. ShanghaiTech Campus (Liu et al., 2018) comprises 13 outdoor scenes, 437 videos, and a mixture of pedestrian, bicycle, and vehicle activity; anomalies (bicycle riding in pedestrian zones, unauthorized vehicles, etc.) are typically defined by a pedestrian–object interaction, making it the natural benchmark for our hypothesis. All skeletons are AlphaPose (Fang et al., 2017) COCO-17 keypoints converted to COCO-18 (synthetic neck) following SeeKer’s preprocessing; we use the publicly released pose JSONs from STG-NF for full reproducibility.

3 METHOD

3.1 BASELINE: SEEKER IN DETAIL

Let a pedestrian track be a sequence of T frames, each containing $V = 18$ COCO-18 2D keypoints. We denote an individual coordinate by $X_{t,n} \in \mathbb{R}^2$, the n -th keypoint of frame t , with $n \in \{1, \dots, V\}$. Let $X_{<t}$ denote all keypoints from frames before t , and $X_{t,<n}$ all keypoints of frame t with index strictly less than n . SeeKer factorizes the joint density of the entire track as

$$p_\theta(X_{1:T}) = \prod_{t=1}^T \prod_{n=1}^V p_\theta(X_{t,n} | X_{t,<n}, X_{<t}). \quad (1)$$

Each conditional is parameterized as a 2D Gaussian $\mathcal{N}(\mu_\theta, \Sigma_\theta)$ whose mean and covariance are produced by a single masked fully-connected network (*MADEPartial*). To guarantee positive-definite covariances, the network outputs the entries of the Cholesky factor $L_\theta \in \mathbb{R}^{2 \times 2}$ such that $\Sigma_\theta^{-1} = L_\theta L_\theta^\top$. The keypoint-level autoregressive constraint is enforced by binary masks on the weight matrices, exactly as in MADE (Germain et al., 2015); SeeKer adapts the masking so that conditioning is causal both *across* time and *within* the keypoint ordering of the current frame.

Training. The model is trained on *normal* pedestrian sequences only, by minimizing the negative log-likelihood of Equation 1.

Anomaly scoring. At inference, each test frame receives a per-keypoint negative log-likelihood. SeeKer aggregates these into a frame-level anomaly score (optionally weighted by AlphaPose detection confidence), then applies temporal Gaussian smoothing. The resulting score is the Mahalanobis distance under the predicted Σ_θ : when the network is confident (small Σ_θ) and the observation deviates from μ_θ , the score is large.

3.2 PROPOSED EXTENSION: CCSKDE

Our contribution is to make the predicted distribution *context-aware*, by introducing a per-frame context vector C_t :

$$p_\theta(X_{t,n} | X_{t,<n}, X_{<t}, C_t). \quad (2)$$

Constructing C_t . For each frame t in the training and test videos, we run YOLOv11 on the raw RGB frame and retain only detections of non-pedestrian classes that are plausible hazards on a campus: *car, truck, bus, motorcycle, bicycle*. For each such detection b , with bounding-box centre (x_b, y_b) , we compute its image-space distance to the target pedestrian’s skeleton centroid $\bar{x}_t$: $d_{t,b} = \|(x_b, y_b) - \bar{x}_t\|_2 / \sqrt{H^2 + W^2}$, normalized by the image diagonal so the quantity is scene-scale-independent. We aggregate to a fixed-dimension vector by taking, per class, the inverse minimum distance and the count of objects within a fixed radius:

$$C_t = [1/d_{c_1}^{\min}, n_{c_1}^{\leq r}, 1/d_{c_2}^{\min}, n_{c_2}^{\leq r}, \dots] \in \mathbb{R}^{2K}, \quad (3)$$

where K is the number of hazard classes. Frames with no relevant detection receive an all-zero vector. This design has three properties we care about: (i) it is fixed-dimension regardless of how many objects are in the scene, (ii) its magnitude grows monotonically with proximity to a hazard, and (iii) it is computed once offline, so it adds no train-time inference cost on the YOLO side.

Injection into the autoregressive backbone. Naively concatenating C_t to the MADE input would interfere with the keypoint mask. We instead treat C_t as an *unconditional shortcut input*: its entries are appended to the input vector and assigned a constant “zero-degree” label in MADE’s connectivity scheme. This construction guarantees that every output neuron may attend to every entry of C_t , while the existing keypoint-to-keypoint causality is preserved. Concretely, only one row block of

MADE’s mask matrices is altered (the columns corresponding to C_t are set to all-ones); no parameter is removed from the existing model. A diagram of the modified architecture will accompany the final report.

Effect on the predicted distribution. Because both μ_θ and L_θ become functions of C_t , the network can in principle learn two complementary effects: (a) shift the predicted mean to anticipate, e.g., evasive sidestepping near a vehicle during normal training behavior, and (b) shrink the predicted covariance in high-context regions, since fewer kinematic outcomes are normal there. Mechanism (b) is the one we particularly care about for hazard detection: under a shrunken Σ_θ , even a small kinematic deviation from μ_θ produces a large Mahalanobis score, exactly when we want the detector to fire. Our ablation will isolate this effect.

3.3 TRAINING AND EVALUATION PROTOCOL

We pre-extract C_t for every frame of ShanghaiTech once, offline, and cache the result as `.npy` tensors keyed by `<scene>_<clip>`. CCSKDE is then trained on normal training sequences with the same NLL objective as SeeKer, with the only change being the augmented input. At test time, scoring is unchanged: per-keypoint NLL, Gaussian-smoothed, confidence-weighted. We compare frame-level AUROC and AP against the vanilla SeeKer baseline trained on the identical pose data, and report a “shuffled-context” negative control that permutes C_t across frames at training time — this verifies any gains arise from the spatio-temporal alignment of C_t rather than from extra parameters.

4 EXPERIMENTAL SETTINGS AND PRELIMINARY RESULTS

4.1 DATASET

We use the ShanghaiTech Campus dataset (Liu et al., 2018): 13 outdoor scenes, 330 normal training videos and 107 test videos containing 130 labelled abnormal events. Anomalies include riding bicycles or driving vehicles in pedestrian areas — precisely the type of context-dependent hazard our method targets. Pose data are the AlphaPose-tracked JSONs released with STG-NF (Hirschorn & Avidan, 2023), which match the format expected by SeeKer. Frame-level ground-truth masks are distributed alongside.

4.2 IMPLEMENTATION ENVIRONMENT

All experiments are run in a Python 3.12 virtual environment with PyTorch 2.11.0 on the Apple Silicon MPS backend. Object detection uses the Ultralytics package (v8.4.46) with YOLOv11. The codebase is organized as follows:

- `seeker/` — a vendored, read-only clone of the official SeeKer release (Delić et al., 2025) (github.com/adelic99/seeker), licensed MIT.
- `ccskde/` — our extension package, containing the context-extraction pipeline (`ccskde.context`), the context-conditioned MADE variant (`ccskde.models`), and dataset wrappers that pair pose tensors with C_t (`ccskde.data`).
- `scripts/` — launchers for baseline reproduction and CCSKDE ablation runs.

We applied a small set of unavoidable portability patches to the upstream `seeker/` package (five hardcoded `'cuda'` references replaced with the device-agnostic `args.device`; one upstream attribute typo, `args.dropout` vs. the registered `args.droppout`, fixed along the way). No model-level edits were made to the baseline.

4.3 HYPERPARAMETERS

For baseline reproduction we follow the SeeKer paper’s published settings on ShanghaiTech: segment length $T = 24$ frames, segment stride 1, batch size 1024 (reduced to 256 for our preliminary smoke run on MPS), 10 training epochs, Adam optimizer with learning rate 5×10^{-4} , single-layer

MADEPartial with expansion factor 1. Anomaly scores are smoothed temporally with a Gaussian filter of σ matched to the paper’s choice. For CCSKDE the hyperparameters are inherited unchanged; only the input dimension grows by $|C_t| = 2K$.

4.4 EVALUATION METRICS

We report frame-level Area Under the ROC curve (AUROC) and Average Precision (AP), the two standard metrics for VAD on ShanghaiTech. The SeeKer paper reports a frame-level AUROC of 85.5% on ShanghaiTech; this is our primary reproduction target and the baseline against which any gain from C_t will be measured.

4.5 PRELIMINARY RESULTS

At the time of this progress report we have verified the end-to-end pipeline and produced a first preliminary number. The vendored SeeKer codebase is patched for device portability and successfully constructs the ShanghaiTech pose dataset on MPS hardware (full 437-video pose split processed in ≈ 3.5 minutes, producing the expected $(N, C=3, T=24, V=18)$ tensor). A one-epoch training run with `batch_size=256` on Apple Silicon MPS completes the full forward/backward pass through MADEPartial, runs the temporal Gaussian-smoothed evaluation, and reports **AUROC** = 0.744, **Average Precision** = 0.647, with $\text{FPR@95} = 0.685$. The paper reports 0.855 AUROC at 10 epochs / batch 1024 on CUDA, so 0.744 after a single epoch on MPS is consistent with a healthy training trajectory; the loss decreases from ~ 21 at epoch start into the negative regime expected of a Gaussian log-density. End-to-end wall-clock is ≈ 20 minutes per epoch on this hardware, so the full 10-epoch reproduction targeting 0.855 AUROC is tractable in a single training session.

Sibling package `ccskde/` is scaffolded with stubs for context extraction, context-conditioned model variants, and pose+context dataset wrappers; the YOLOv11-based C_t pass will run as a one-off preprocessing step once baseline reproduction is confirmed. The final report will include the reproduced baseline, the CCSKDE result, the with/without- C_t ablation, and the shuffled-context negative control described in Section 3.3.

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